

Alapan Das

📍 Paris ✉ alapan.das@phys.ens.fr in LinkedIn

Employement

Laboratoire de Physique de l'École Normale Supérieure, ENS-PSL, Paris,
Doctorat en physique

Sept 2025 – present

Advisor: Prof. Adam Nahum

Thesis title: Unusual phase transitions due to quantum measurements

Education

Tata Institute of Fundamental Research, Mumbai

Aug 2022 – July 2025

Master of Science in Theoretical Physics

- GPA: 8.4/10

Thesis title: Ising model on hyperbolic lattices (Grade: 9/10)

Advisor: Prof. Kedar Damle.

Visva Bharati University, Santiniketan

July 2019 – Aug 2022

Bachelor of Science, Physics

- GPA: 8.55/10

Birbhum Zilla School, Suri

2009 – 2019

10+2, Science

- Grade: 94%

Publications and pre-prints

Bayesian phase transition for critical Ising model

- We study the Bayesian measurement-induced transition for the critical Ising model in $6 - \epsilon$ and 2 dimensions and discover an enlarged (or emergent) symmetry similar to the one found along the Nishimori line. ([arXiv:2604.23346](https://arxiv.org/abs/2604.23346) [↗](#), with Prof. Kay Jörg Wiese and Prof. Adam Nahum)

Projects

Ising model on Hyperbolic Lattices (Master's thesis)

Jan 2025-June 2025

- Studied the free fermionic spectrum of the Ising model on the hyperbolic lattices. In this project, I constructed a dimer mapping for the hyperbolic lattice that respects the Fuchsian group symmetry. I employed the generalised Bloch theorem for non-Abelian Fuchsian groups to study the free fermionic spectrum of the system to understand several properties of this system. (*Manuscript under preparation, with Prof. Kedar Damle*)

Free Fermionic Representation of RBIM

Feb 2024-Dec 2024

- We constructed a dimer mapping with a real Hamiltonian for random bond Ising models in two dimensions. From this, we studied the low-energy modes of this free fermionic model, which are introduced by frustration. We connected the frustration to a monomer travelling problem and found a lower bound for the cumulative distribution function of these modes, which helps us understand Griffith's phase better. (*Manuscript under preparation, with Prof. Kedar Damle*)

Academic achievements

Joint Entrance Screening Test, Physics *Mar 2022*
Score-74/100, All India Rank - 1

IIT-IISc Joint Admission test for Master's, Physics *Feb 2022*
Score-89.67/100, All India Rank - 2

Jagadish Bose National Science Talent Search (JBNSTS), Kolkata *2019 & 2017*
Senior and Junior Scholar

DST-Inspire scholarship *2019-2022*
Govt. of India

Conferences and Workshops

School on Frustrated Quantum Magnets *Oct 28 - Nov 8, 2024*
École de Physique des Houches, Les Houches, France

Stability of Quantum Matters In and Out of Equilibrium at Various Scales *15-24 Jan, 2024*
ICTS-TIFR, Bangalore

IISER-JBNSTS workshop, IISER Kolkata *July 2018*
Workshop organised for JBNSTS junior scholars

Work Experience

Teaching assistant for Advanced Quantum Mechanics course *Aug-Dec 2024*
Instructor: Prof. Rajdeep Sensarma

Teaching assistant for Mathematical methods course *Aug-Dec 2023*
Instructor: Prof. Shravan Hanasoge

Skills

C++, Python, Mathematica, Inkscape

Extracurricular activities

Secretary of TIFR movie club *Aug 2024 - May 2025*
I was a co-secretary of the movie club of the TIFR student society